

FACULTY OF ENGINEERING

GA ASSIGNMENT

INGB417

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1. Academic Integrity

I conducted this project with full academic integrity, ensuring that all work is original and completed in accordance with the ethical standards and guidelines of the North-West University and its Faculty of Engineering.

2. Executive Summary

An industrial engineers has been appointed as facility designers to design a new facility for TIPTOP TABLES. The facility has to be designed from scratch. The facility is required to make lightweight aluminium outdoor tables and accompanying chairs with rattan backing. A 20% expansion must also be considered within the next few years. The facility must meet a demand of 5 000 units monthly.

3. Background

TIPTOP TABLES is a South African manufacturer that has supplied outdoor tables and chairs for over 25 years. Due to increasing demand, especially for a lightweight, configurable aluminium outdoor table set (1 table + 4 chairs), the company now forecasts monthly sales of up to 5 000 sets. The existing facility cannot meet this demand, prompting the need for a new purpose-built manufacturing facility focused solely on this product line. The company requires a comprehensive facility design that incorporates production processes, storage, shipping, receiving and potential for future expansion (20% spare capacity), developed through a systematic engineering design approach.

4. Design Problem and Objectives

The core design problem is to develop a fully functional and scalable facility layout that meets the forecasted production demand of 5 000 outdoor table sets per month. This must include:

- Efficient space allocation for all production and non-production functions
- Logical material and product flow throughout the facility
- Compliance with safety and operational constraints (e.g., isolated welding areas, safety zones for robots)
- Integrated support areas including offices, employee facilities and logistics zones
- Alignment with inventory and delivery policies, ensuring adequate storage capacity
- Flexible layout with provision for future capacity expansion

5. Design Requirements

- Meet monthly demand of 5000 units
- 20% expansion space
- Isolated areas for welding
- Safety space around where robots are working
- Should conform to activity relationship chart
- Optimise facility layout
- Sufficient external interface
- Logical flow of material
- Access to roads
- Should be easy to implement or construct

6. Design Assumptions and Constraints

6.1. Process Flow

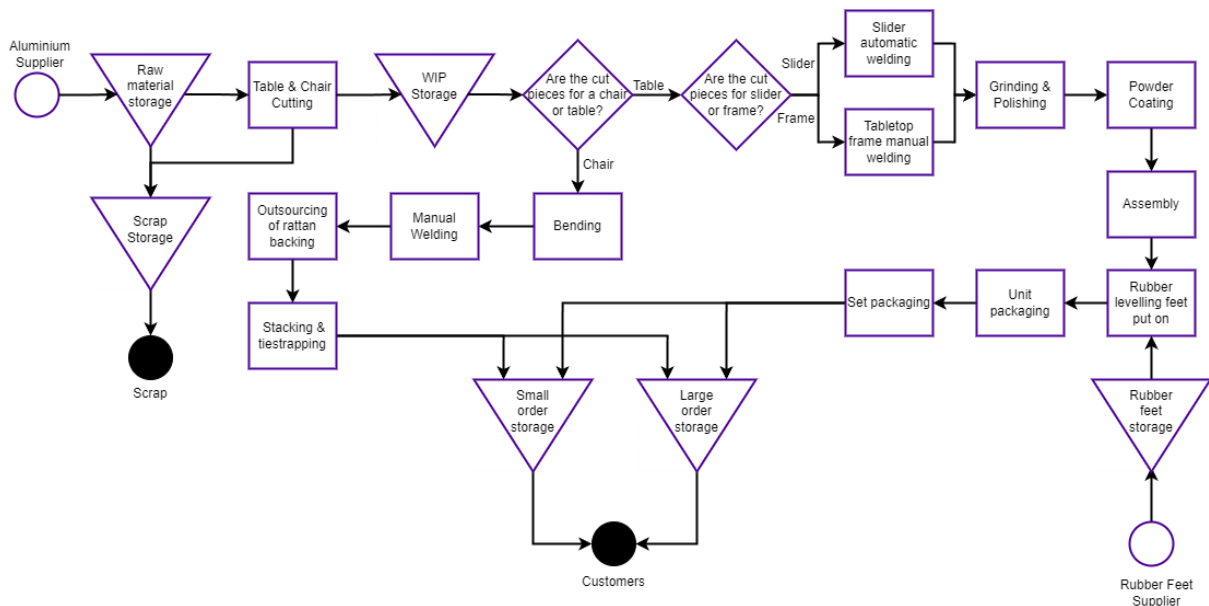
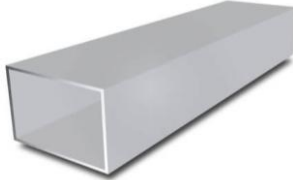
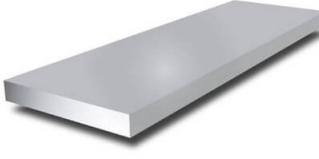


Figure 6.1: High-level Process Flow Chart

6.2. Assumptions & Constraints

- All bars and tubes are assumed to be 6m in length before being cut
- A unit is one set, including 1 tabletop, four legs and four chairs
- Small orders are 20% of total orders
- 20 working days per month
- Productivity is 7.5hrs daily
- Setup time and breakdowns take up 0.5hrs daily
- 5% scrap rate from aluminium delivered by supplier

Table 6.1: Raw Material Item, Dimensions and Use

Item & Dimensions	Image	Use
Rectangular Tubes (40mmx20mmx1,2mm)	 Figure 6.2: Rectangular Tube	Frame for tabletop Sliding structure
Flat Bars (101,6mmx9,52mm)	 Figure 6.3: Flat Bar	Tabletop

Square Tubes (60mmx60mm)	 <i>Figure 6.4: Square Tube</i>	Table legs
Round Tubes (25,4mmx1,6mm)	 <i>Figure 6.5: Round Tube</i>	Chairs
Rubber Levelling Feet	 <i>Figure 6.6: Levelling Feet</i>	Levelling Tables

The product dimensions can be seen in table 6.2 below. The length and number of slats used for the second tabletop is calculated by working out the length difference between the large and small configuration of tables and then using that ratio to determine the amount of slats (flat bars) needed. Product images can be seen in figures 6.7 – 6.10.

Table 6.2: Product Description

Item	Dimensions (l*w*h)
Tabletop 1	1.52m*1m*0.04m 15 flat bars
Tabletop 2	0.96m*1m*0.04m 10 flatbars
Slider insert	0.94m*1m*0.04m
Legs	0.06m*0.06m*0.7m
Chair	0.55m*0.59m*0.76m



Figure 6.7: Finished Expanded Table

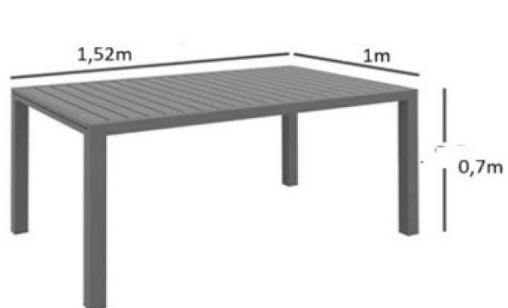


Figure 6.8: Small Configuration

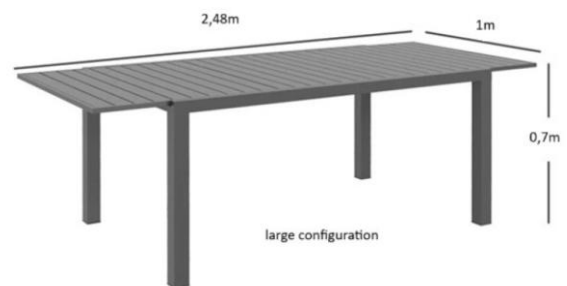


Figure 6.9: Large Configuration



Figure 6.10: Chair

7. Detail Design Specifications

7.1. Production Volumes

The daily production volumes are calculated by taking the monthly demand (5000 units) and dividing it by 12 months and then dividing that value with 20 working days per month. This results in a total of 250 units that have to be produced daily. This is a total of 250 sliders, 500 tabletops, 1000 legs and 1000 chairs.

The amount of machines needed per type is calculated by multiplying the number of units per process by the processing time per item and dividing by the minutes per day (450mins). A summary of the amount of machines needed for each process can be seen in table 7.1 below.

Table 7.1: Machines Calculations

Machine	Units	Processing Time (mins)	Productivity per day (mins)	Machines Needed
Cutting	250x tables 1000x chairs	1	450	3
Bending	1000 chairs	3	450	7
Automated Welding	250 slider insert	5	450	3
Manual Welding	500 tabletops 1000 chairs	10 (tabletop) 5 (chair)	450	24
Grinder	500 tabletops	0.5	450	1
Polisher	500 tabletops	0.5	450	1
Powder Coater	500 tabletops 250 slider inserts 1000 legs	1.5 (30sec for 1 st tabletop + 10sec for each 2 nd tabletop, slider insert, legs)	450	1
Packaging	50 sets	5	450	1

7.2. Departments

Raw Material Storage

In order to determine how many 6m bars and tubes of each type have to ordered, the total length of each type for one unit is calculated first. This is done by adding the lengths of all cut pieces up per type of bar or tube and then dividing that by 6m (length of bar/tube). This final value is then rounded up to get the number of bars or tubes needed per type per unit. The final amounts are calculated by multiplying the unit amount by 250 (daily demand). A summary of these calculations can be seen in table 7.2 below.

Raw materials are delivered once daily. Four workers are employed by TIPTOP TABLES to pack the racks and check for defects. Forklifts are used to unload the delivery trucks, after which smaller batches are put in trolleys for easier distribution between the racks.

Table 7.2: Number of 6m bars and tubes needed for daily production

Item	Cut pieces per unit	Bars per unit	Bars per day (250 units)	2-Weekly raw materials with scrap
Rectangular Bars	3x1.52m 7x1m 3x0.96m 2x0.94m	2.836	709	7464
Flat Bars	25x1m	4.168	1042	10969
Square Tubes	4x0.7m	0.5	125	1316
Round Tubes	4x6m	4	1000	10527
Totals	-	11.504	2876	30276

Cantilever racking is used to store the 6m tubes and bars. The racks have a height of 2m, arm length of 0.8m and peg height of 0.3m. It is assumed that each rack has a height of 0.6m and length of 6m. The type of rack that will be used can be seen in figure 7.1 below. The calculations for the number of racks needed can also be seen below in table 7.3. Five of these racks are allocated to the defective tubes and bars received. The workstation layout of one of these racks is depicted below in figure 7.2.



Figure 7.1: Cantilever Rack

Table 7.3: Cantilever Rack Calculations

Item	Bars or Tubes received	Bars/Tubes in a layer	Layers per rack	Racks
Calculation	-	$\frac{\text{Rack width}}{\text{bar/tube width}}$	$\frac{\text{Rack height}}{\text{bar/tube height}}$	$\frac{\text{raw material} + 5\% \text{ scrap}}{6 * \text{layers} * \text{items in layer}}$
Rectangular Bar	7464	20	15	5
Flat Bar	10969	7,874016	3,151261	74
Square Tube	1316	13,333333	5	4
Round Tube	10527	31,49606	11,81102	5
Total	30276	72,70341	34,96228	88

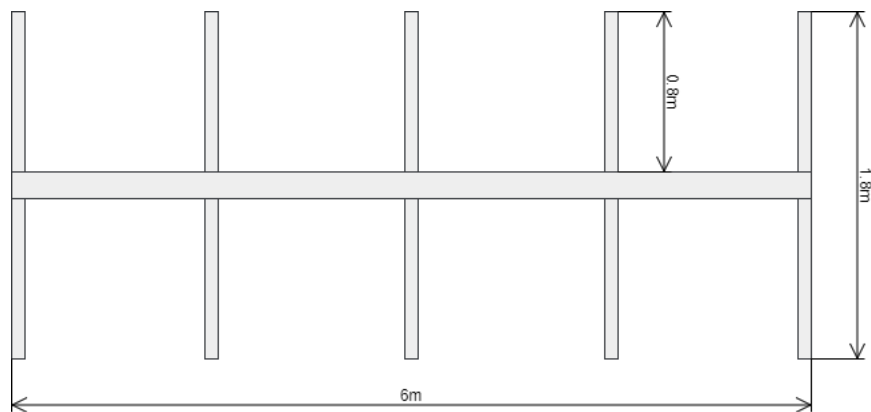


Figure 7.2: Workstation Layout Cantilever Rack

Work-in-Progress

Racks are used to store the bars after they have been cut after which they are then taken to the next work areas. These racks are placed after each workstation to accommodate work-in-progress (WIP) materials. An example of the type of racks used is depicted in figure 7.3 below.



Figure 7.3: Storage Racks

These racks have the dimensions of 2mx0.6mx2m. By calculating the area needed for all the items, the number of racks needed can be found by dividing this area by the shelf space of one rack (area of one shelf multiplied by six shelves). Items are packed in two layers on each rack and thus effectively doubles the area a rack can take. From this it is calculated that a total of 35 racks is needed at each WIP area. 19 of these racks are for the tables and 16 for the chairs. The chair storage is only used at the cutting process and welding process. As pieces are used by the following station as soon as possible, only half of the racks are needed. The calculations can be seen in table 7.4 below.

Table 7.4: Work-in-Progress Space Calculations

Item	Area (m ²)	Quantity	Area Needed (m ²)	Rack Area	Racks Needed
Tabletop 1	1,52	250	380	48	8
Tabletop 2	0,96	250	240	48	5
Legs	0,042	1000	42	48	1
Slider Insert	0,94	250	235	48	5
Chair Cut Pieces	0,1524	5000	762	48	16
Totals					35

Cutting

Pieces are cut in bulk in order to reduce the number of cutoffs from each 6m bar or tube. The following table (table 7.5) indicates how many of which cuts are made as well as the number of bars needed daily when the 5% scrap rate is considered.

Table 7.5: Cutting Calculations

Type of bar/tube	Pieces cut from 6m	6m bars/tubes needed	6m bars/tubes with 5% scrap	Length utilised per 6m tube/bar
Rectangular Bars	3x1,52m + 1x1m	250	263,1578947	5,56
	6x1m	250	263,1578947	6
	6x0,94m	84	88,42105263	5,64
	6x0,96m	125	131,5789474	5,76
Flat Bars	6x1m	1042	1096,842105	6
Square Tubes	8x0,7m	125	131,5789474	5,6
Round Tubes	1x6m	1000	1052,631579	6
Totals	-	2876	3027,368421	-

From table 7.5 above it is calculated that 3 machines are needed to meet demand. The same machine is used for all bars and tubes and once one type is finished being cut, the next type's dimensions are put in and cut. The machine is depicted in figure 7.4 and the workstation layout in figure 7.4 below. The specifications of the machine can be seen in table 7.6 below.



Figure 7.4: Cutting Machine

Table 7.6: Cutting Machine Specifications

Dimensions (l×w×h)	8.55 × 1.3× 2 (m)
Operators per machine	2
Machines Needed	3

The total volume needed for cutoffs is 0.3162m³. The calculations for this is summarised in table 7.7 below. The cutoff length per bar is multiplied with the amount of bars or tubes of that type, the width and height of that type of bar or tube.

Table 7.7: Cutoff volume calculations

Pieces x length (in 6m)	6m Bars/Tubes	Cutoff length	Cutoff pieces volume (m ³)
3x1,52m + 1x1m	250	0,44	0,088
6x1m	250	0	0
6x0,94m	84	0,36	0,024192
6x0,96m	125	0,24	0,024
6x1m	1042	0	0
8x0,7m	125	0,4	0,18
1x6m	1000	0	0
Total	-	-	0,3162

One bin of volume 0.11m³ is placed at each cutting machine for the cutoffs. The bin is depicted in figure 7.5 below. The cutoffs in these bins are collected daily and taken away.



Figure 7.5: Cutoff bin

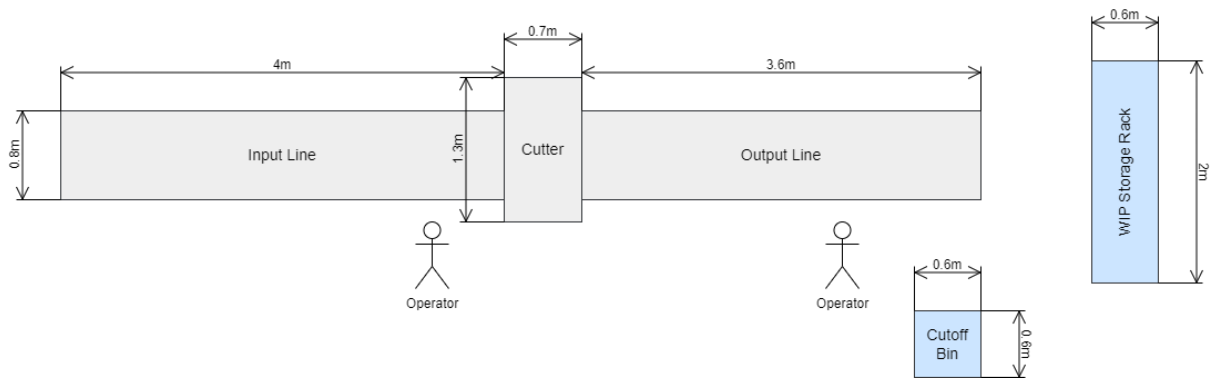


Figure 7.6: Cutting Workstation Layout

Bending

All cut round bar pieces are moved to the bending machines from the cutting area. From table 7.1 above it is calculated that 7 machines are needed. The bending machine is depicted in figure 7.7 and its workstation layout in figure 7.8 below. The specifications of the machine can be seen in table 7.8 below. There are no racks used at the WIP area of the bending as it is easier and more space efficient to stack the chairs in the allocated WIP area. Three stacks of chairs can be put next to each other and moved to the rattan backing area.



Figure 7.7: Bending Machine

Table 7.8: Bending Machine Specifications

Dimensions (l×w×h)	2.6 × 0.6 × 1.2 (m)
Operators per machine	1
Machines Needed	7

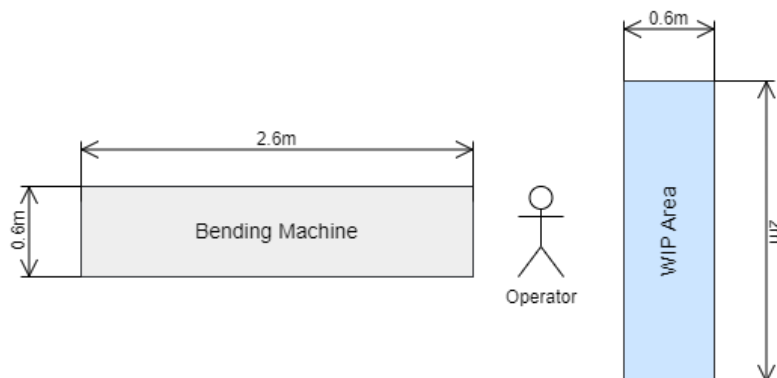


Figure 7.8: Bending Workstation Layout

Welding

After bending, the pieces are taken to the welding area. The rest of the cut pieces to be used for the tables are brought from the cutting area to the welding area. Automated welding machines are used to make the slider insert while manual welding machines are used to make the tabletops and chairs.

From table 7.1 above it is calculated that 3 automated welding machines and 24 manual welding machines are needed. These machines and their specifications are depicted in figures 7.9 & 7.10 and table 7.9 below.

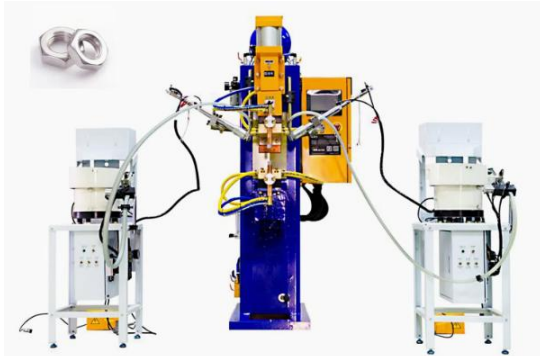


Figure 7.9: Automated Welding Machine



Figure 7.10: Manual Welding Machine

Table 7.9: Welding Machines Specifications

	Automated	Manual
Dimensions (l×w×h)	1.7 × 0.8 × 1.6 (m)	0.8 × 1.4 × 1.4 (m)
Operators per machine	1	1
Machines Needed	3	24

Both workstations also have additional tables at which the machines are used. These tables are big enough for one tabletop to be welded at a time. The workstation layouts are depicted in figures 7.11 and 7.12. A 1m safety space is added all around the workstation of the automated welding robots.

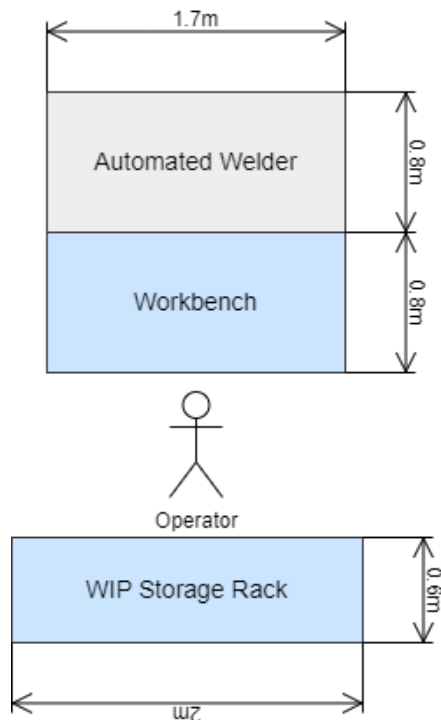


Figure 7.11: Automated Welding Workstation Layout

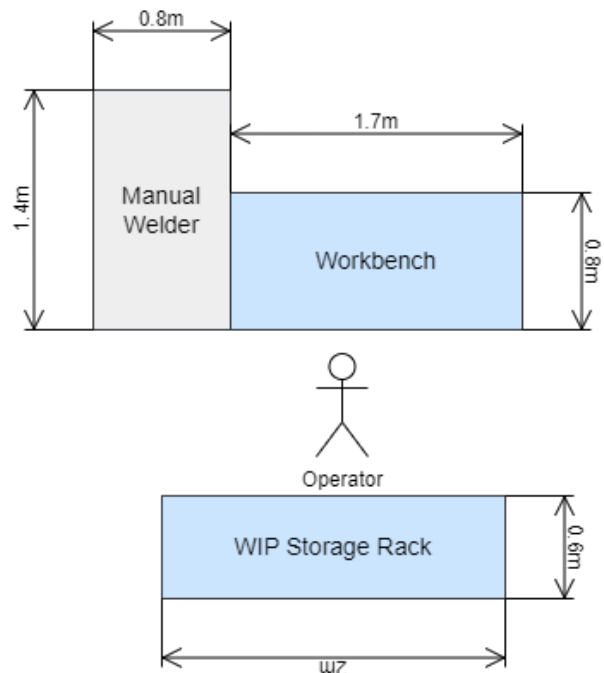


Figure 7.12: Manual Welding Workstation Layout

Grinding & Polishing

The table's welded pieces are taken to the grinding & polishing area. The welded chairs are taken to the rattan backing storage area. Handheld grinders and polishers are used to grind excess welding away on the tables and to polish the pieces before they are ready for the powder coating.

From table 7.1 above it is calculated that 1 grinder and 1 polisher are needed. These machines and their specifications are depicted in figures 7.13 & 7.14 and table 7.10 below. The amount of grinder discs needed daily is also indicated. This was calculated by taking the average life span of a grinder disc (30min of continuous grinding) and dividing the productivity minutes (450min) by that. This results in 15 discs per grinder per day. The discs needed for the polisher is calculated in the same way. The average polisher disc lifetime is 40mins and this results in 12 disks needed per day per polisher. The polishers and grinders' dimensions are small, but the workstation must be big enough to accommodate one tabletop at a time. The workstation layouts can also be seen below in figure 7.15.



Figure 7.13: Handheld Grinder



Figure 7.14: Handheld Polisher

Table 7.10: Grinding & Polishing Specifications

	Grinder	Polisher
Operators per machine	1	1
Machines Needed	2	2
Discs Needed	15	12

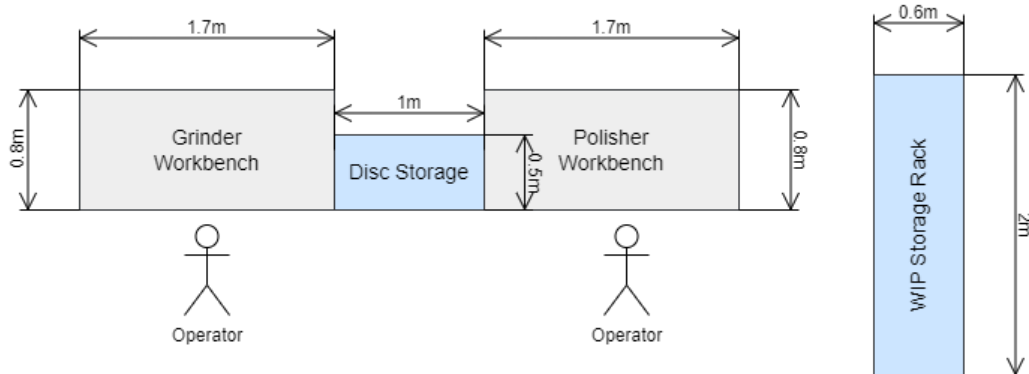


Figure 7.15: Grinding & Polishing Workstation Layout

Powder Coating

From the grinding and polishing areas, the table's welded pieces are taken to the powder coating machine. Here all seven items used to make a table is sent through the powder coater with 10sec intervals in-between each item. It is thus assumed that it takes 1.5mins for all items to go through the powder coater. This consists of 30secs for the first item plus an additional 10secs for the next items. Only one powder coater is needed to meet demand.

The powder coating machine is depicted in figure 7.16 below. The specifications can be seen in table 7.11 below. The workstation layout is depicted in figure 7.17.

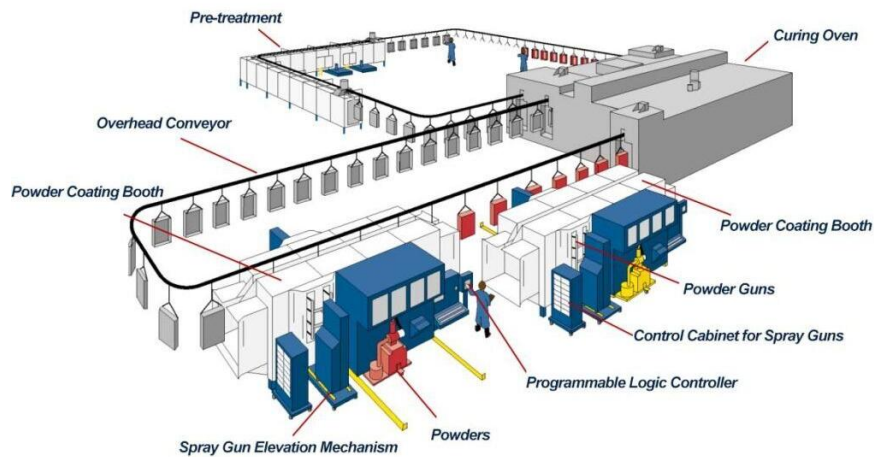


Figure 7.16: Powder Coating Machine

Table 7.11: Powder Coating Machine Specifications

Dimensions (l×w×h)	30 × 18 × 3 (m)
Operators per machine	6
Machines Needed	1

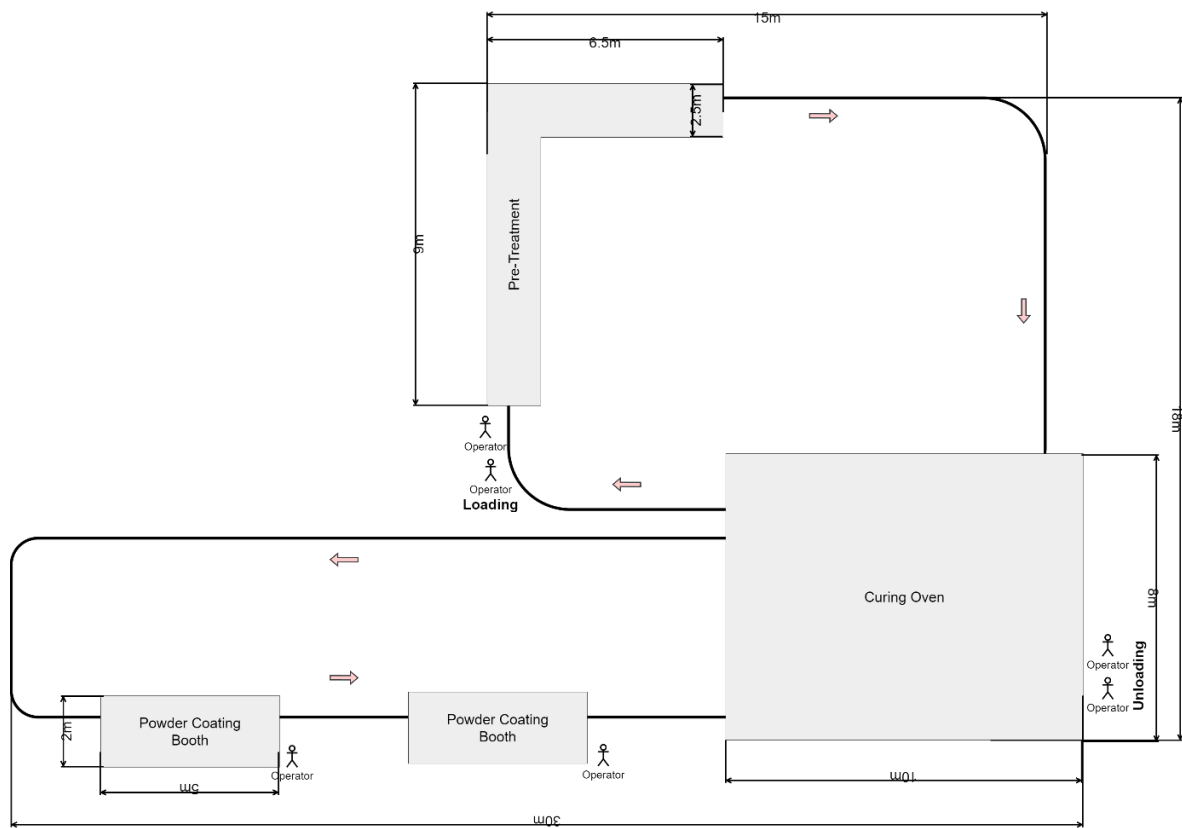


Figure 7.17: Powder Coating Workstation Layout

Assembly

No machinery is used with the assembly of the tables. Workbenches are however required for the tables to be assembled on. From table 7.1 above it can be seen that 3 workbenches are necessary to meet daily demand. An example of the type of workbench to be used can be seen in figure 7.18 below. The specifications are also shown below in table 7.12. WIP storage racks are placed between the powder coater and the workstations to account for bottlenecks that might occur as the powder coater is

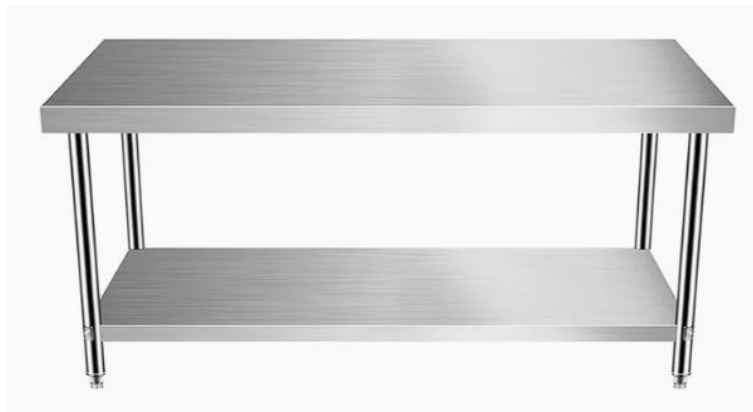


Figure 7.18: Assembly Workbench

Table 7.12: Assembly Workbench Specifications

Dimensions (l×w×h)	1.7 × 0.8 × 0.7 (m)
Operators per workbench	3
Workbenches Needed	3

Rubber levelling feet are placed on the bottom of all the table legs. These feet fit snugly into the square tubing used for the legs. After adding the rubber feet and checking that the legs fit, the legs are removed. This is done to make the packaging process easier. From this area two workers pack five sets of tables on a trolley and move it to the packaging area. An example of the type of trolley used is seen below in Figure 7.19.



Figure 7.19: Trolley

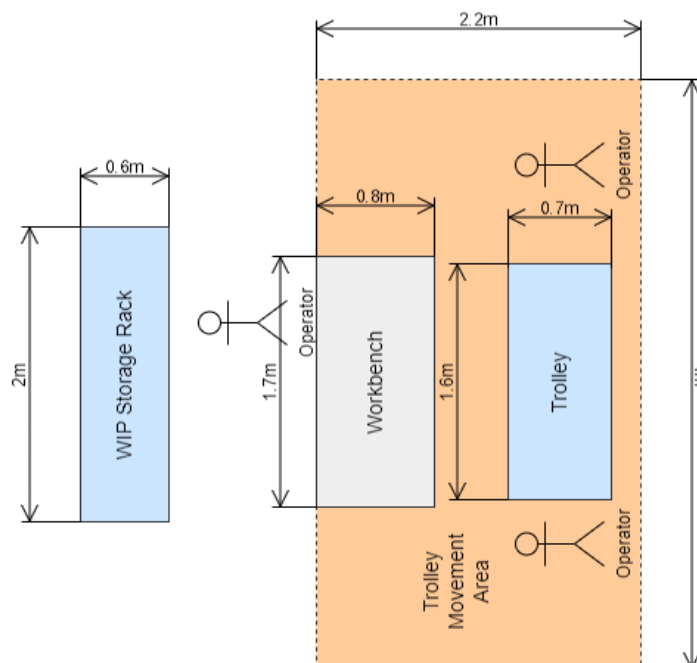


Figure 7.20: Assembly Workstation Layout

Rattan Backing Chair Storage

Enough storage space has to be allotted for one day's chairs with backing and without backing (2000 chairs). Chairs without rattan backing are swapped with chairs with backing everyday by the outsourced people. The chairs are placed on dollies with wheels to ease their movement, since 20 chairs are stacked on top of one another. This means that 100 dollies are needed to accommodate both backed and unbacked chairs. One stack of 20 chairs takes up a floor area of 0.48m^2 ($0.8\text{m} \times 0.6\text{m}$). Multiplying this by 100 dollies, it can be seen that rounded up, 50m^2 are needed for the storage area. Two employees receive and move the finished and unfinished stacks to and from the trucks. Another two move the finished chairs to the finished goods storage area.

The dolly with wheel being used is depicted in figure 7.21 below. The height of the dolly is 0.2m. Using this, the height of one chair (0.76m) and the added height of each additionally stacked chair (0.1m), it can be calculated that the height of a stack of 20 chairs plus the dolly is 2.86m. The finished chairs are tie strapped when received back, ready for delivery.



Figure 7.21: Dolly with wheels

Packaging

The de-assembled tables and chairs are taken to the packaging area. A plastic wrap dispenser is used to package a table and legs together after which 5 sets of tables and legs are then placed in a carton box with dimensions $1.55\text{m} \times 3.25\text{m} \times 1\text{m}$. Unfolded boxes (50) boxes are stacked and stores next to the machine for each day 10 pallets are also stored here to stack 5 boxes on each. Two workers stack these boxes on a forklift and forklifts are used to move it from here to the finished goods area. From table 7.1 above it is calculated that 1 machine is needed. The machine and its specifications are depicted below in figure 7.22 and the workstation layout in figure 7.23 and table 7.13.



Figure 7.22: Plastic Wrap Dispenser

Table 7.13: Plastic Wrap Dispenser Specifications

Dimensions (l×w×h)	1.2 × 1.8 × 2.2 (m)
Operators per machine	4
Machines Needed	3

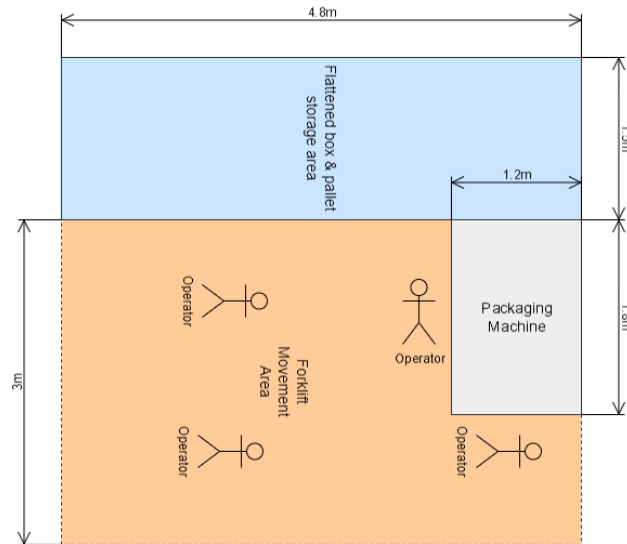


Figure 7.23: Packaging Workstation Layout

Finished Goods Storage

The dimensions of a box with 5 sets of tables are 1.55m x 3.25m x 1m. As there are 5 sets of tables in a box, the total amount of boxes stored for one month is 1000. Two forklifts will be used to move these boxes to the finished goods storage area. Another two will be used to assist on the loading docks. The boxes are stacked 5 on top of one each other. This results in 200 stacks with a floor space of 5.1m² each. Chairs are moved on their dollies by two workers from the rattan backing area to the finished goods storage. 20 000 chairs are stored in this area in a 1000 stacks of 20 chairs each with a floor area of 0.48m² each.

It is assumed that small orders make up 10% of total orders. These orders are removed from the carton boxes and tie strapped stacks as customers come to collect them. Two workers are employed to manage the finished goods storage and dispatching of the items. The final quality inspection is also done at these areas.

Supporting Facilities

Supporting facilities inside the facility include offices, reception, bathrooms, cafeteria and breakroom and locker rooms. Outside of the facility, parking and security gates and checks are considered. The following areas and dimensions as can be seen in table 7.14 below are allocated to these supporting facilities. All of these areas are based on various different national and international standards like SANS 10400.

Table 7.14: Supporting Facilities

Facility	Quantity	Area per item (m ²)
Executive Office	1	12–20
Private Office	2	8–12
Open-plan office	1	4-5 per person
Reception	1	8–15
Meeting Room	1	15–25
File storage	1	4–10
Office Bathrooms	2	6–8
Operator Bathroom & Locker Room	1	70-90
Cafeteria	1	95-105
Parking	100	12.5
Security Checks	3	9

7.3. Distribution of large orders

To determine the needs in terms of trucks, the total floorspace that the tables and chairs take up (calculated above) is divided by the available floor space in a standard 12m high cube truck. A truck of this size has a useable floorspace of 28 m². Dividing this by the 0.48m² one stack of chairs take up, it is found that one truck can take 58 stacks of chairs. It is decided to only transport 40 stacks in a truck and thus two trucks are needed daily for distribution purposes. For the outsourcing of the rattan backing another two trucks are needed as 1000 chairs are taken to the rattan backers and another 1000 is brought back in the same truck.

For the tables, the floorspace needed is 5m² per box. When packing the boxes in two layers, it is calculated that 10 boxes can be shipped per truck. Four trucks are thus needed to meet the daily 80% large order demand.

The following Activity Relationship Chart in figure 8.1 below is setup by using the specifications and limitations by management as well as observing the current manufacturing process.



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Table 8.1: Department Dimensions

Department	Area of one machine (m²)	Machines	Total Area (m²)	Expanded Area (m²)
Raw Material Storage	21,87613636	88	1925,1	2310,12
Cutting	60,75	3	182,25	218,7
Bending	13,6125	7	95,2875	114,345
Manual Welding	14	24	336	403,2
Automated Welding	20.8	3	62.4	74,88
Grinding & Polishing	32	2	64	76,8
Powder Coating	693	1	693	831,6
Assembly	16,8625	3	50,5875	60,705
Rattan Backing	0,48	100	75	90
Loading Area for Rattan Chairs	24	1	24	28,8
Packaging	27,4725	3	82,4175	98,901
Finished Tables	11,27	200	2254	2704,8
Finished Chairs	1,056	1000	1056	1267,2
Forklifts	30,8	1	30.8	36,96
Offices	62	1	62	74,4
Office Bathrooms	16	2	32	38,4
Operator Bathroom & Locker Room	96	1	96	115,2
Cafeteria	206	1	206	247,2
Parking	12,5	100	1250	1500
Totals	-	-	8577	10 293

8.1. Alternative 1

For this alternative, the focus was to keep to the areas and dimensions discussed above in table 8.1. Only a few of the departments' dimensions were adjusted to allow for a more U-shaped design of the building. The flow starts at the raw material loading bays and storage. From there it moves to the various departments as manufacturing process goes until it reaches the finished goods areas. The supporting facilities are placed towards the middle of the facility to allow for easy access.

One of the main issues with this design is the use of corridors for the movement of chairs from the bending department to the rattan backing storage area, where it is then transported through another corridor to reach the loading bay. It is also difficult to reach the finished table storage area from reception. When people come to pick up their small orders they will have to drive from reception to the storage area to collect the items. An employee from reception will have to either drive with them or walk to that area. This design takes up 14 415m² without expansion and with 20% expansion 17 298m².

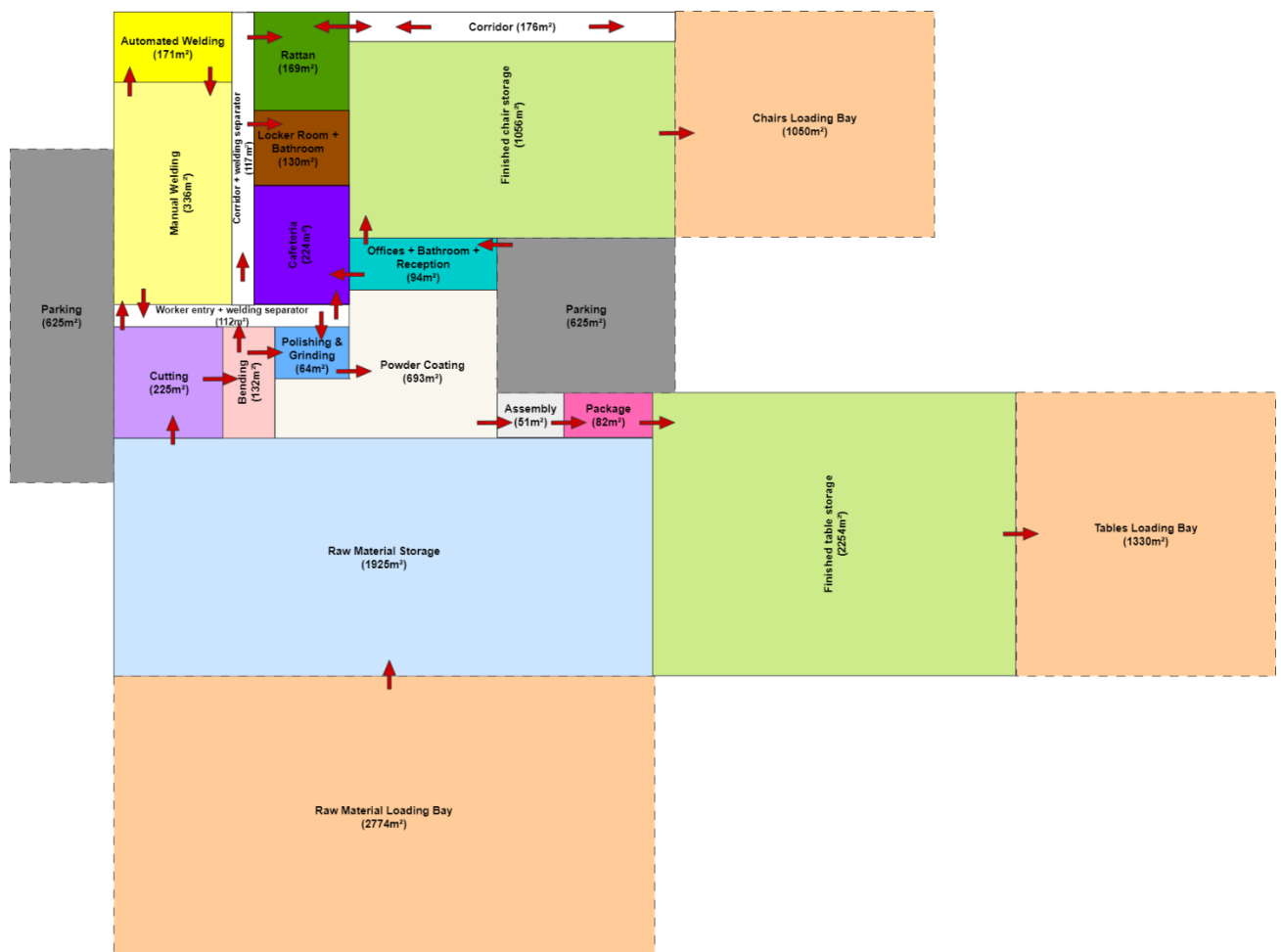


Figure 8.2: Alternative Block Layout 1

8.2. Alternative 2

This design is a bit more compact than alternative 1 in the sense that there is less unused space outside. On the inside however, most of the departments' size is increased a bit to accommodate for better flow. Especially at the bending department where welded materials for the tables are moved through the area to the polishing & grinding area. The finished goods areas are also put together here, making it easier for both small and large order dispatching. The reception area can also directly go into the finished goods storage area. The rattan backing storage area also has its own loading bay, contrasting alternative 1. Workers also have easier access from the parking area to the locker room and offices.

The shape of the building itself is a bit less modular than alternative 1, but better than that of alternative 3. It would still allow for relatively easy construction. One of the small disadvantages of this layout is that a corridor is used again to move chairs from the rattan backing area to the finished goods storage. This corridor is shorter than the one in alternative 1 and is only used for half the amount of chairs (finished chairs). This design takes up 15 224m² without expansion and with 20% expansion 18 293m².

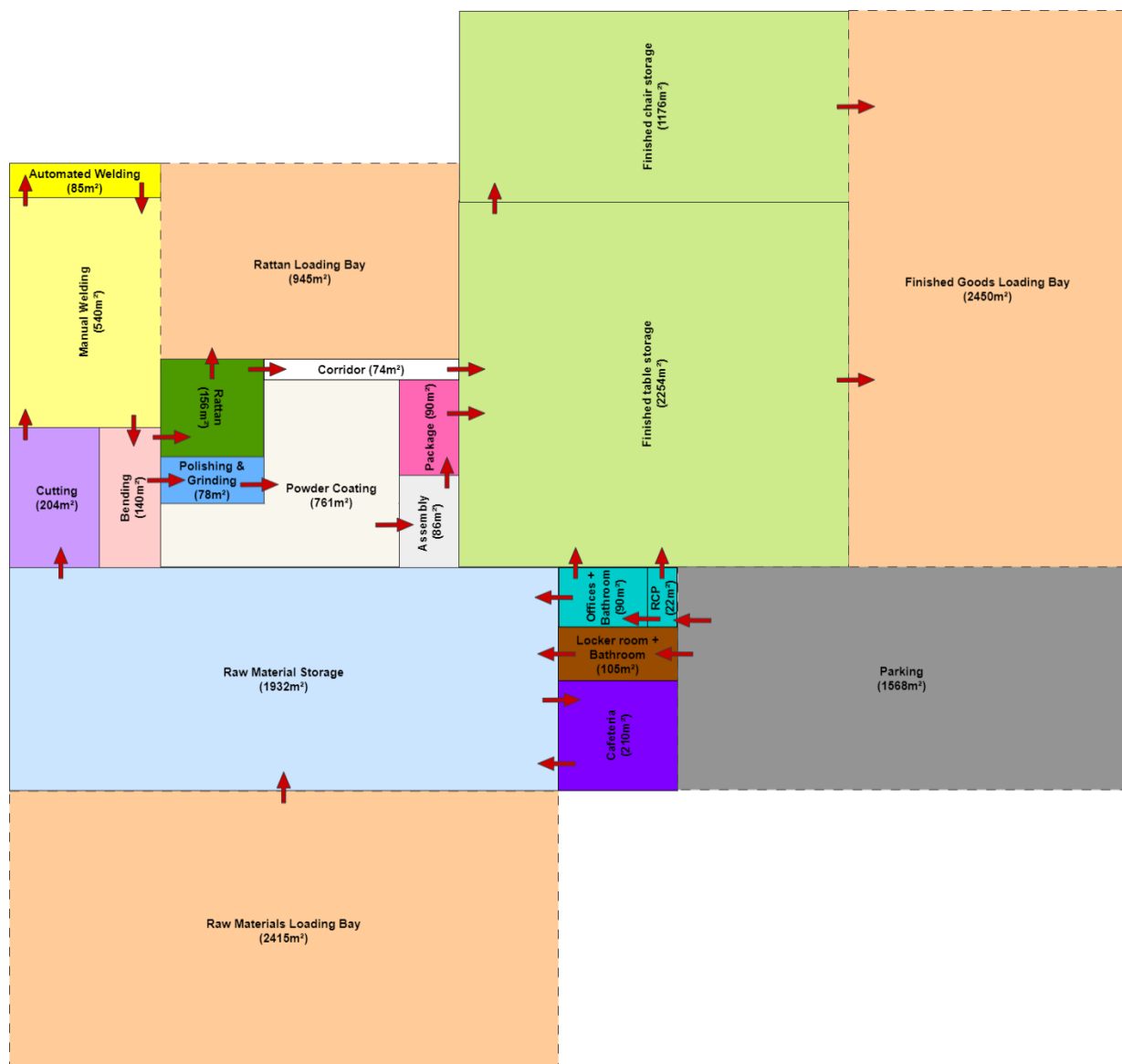


Figure 8.3: Alternative Block Layout 2

8.3. Alternative 3

This layout has by far the most complex shape. This was done in order to try and split the operations between chair and table as far as possible and by putting the common departments in the middle of the design. The flow of this design is a bit simpler due to this design choice, but the odd shape outweighs the small improvement the flow has compared to other alternatives above.

Similarly to alternative 1, in this design there is no direct path for employees in reception (who manages the small orders) to get to the finished chair storage area. Another issue is that the rattan backed and unbacked chairs have to move through the finished goods storage to be loaded and unloaded from trucks. The welding area is also in the middle of the layout, which although it is shut off with walls from the rest of the areas, is not ideal. This design takes up 15 829m² without expansion and with 20% expansion 18 995m².

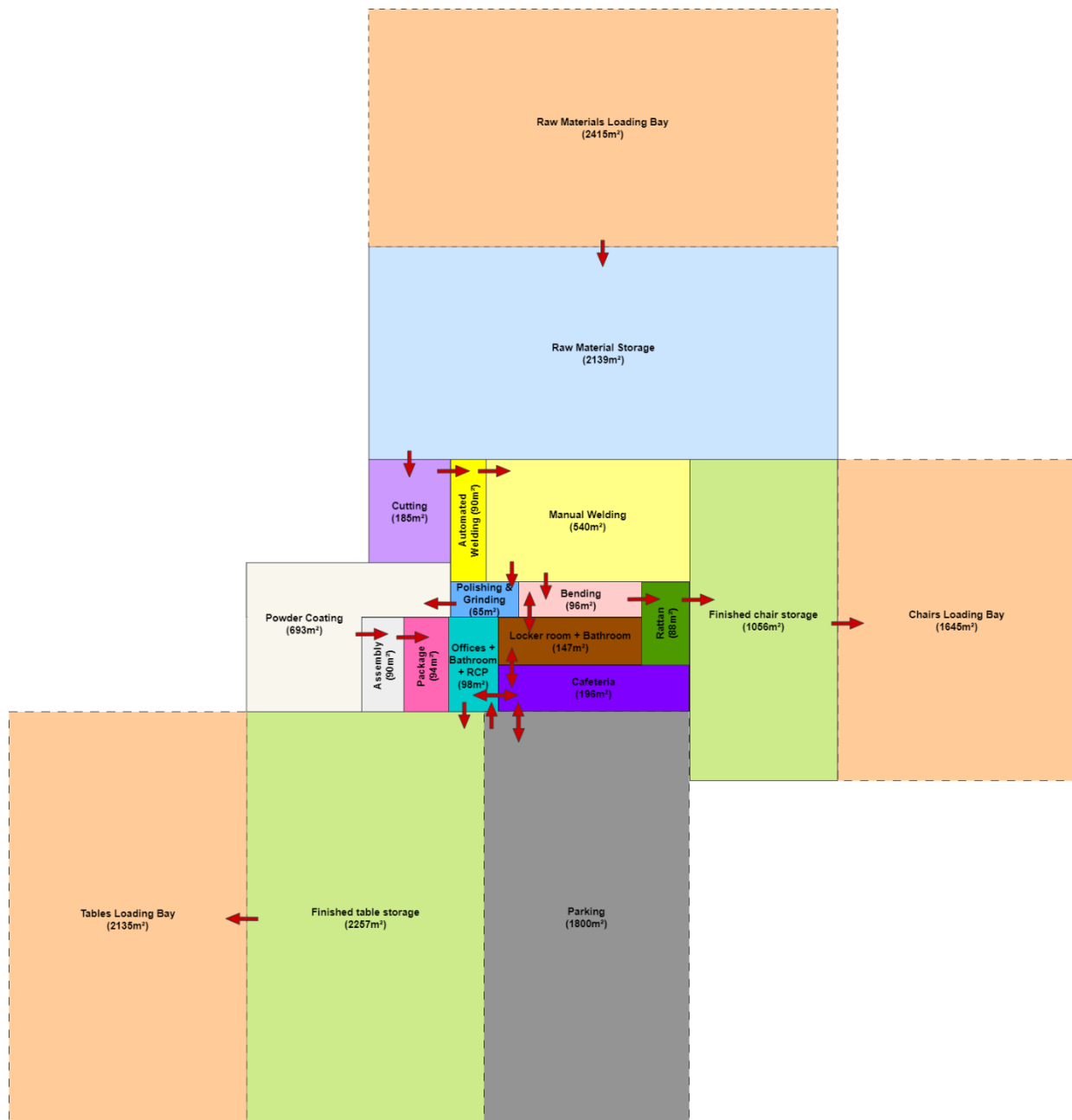


Figure 8.4: Alternative Block Layout 3

8.4. Alternative Evaluation

A weighted decision matrix is set up in table 8.2 below to score and evaluate the different alternatives explained and shown above. The matrix is setup with the design requirements as criteria. Each alternative is then scored out of ten based on the criteria. This score is then multiplied by the weights. The weights add up to one. The alternative design with the highest score is selected to base the MFP on.

Table 8.2: Alternative Evaluation

Criteria	Weight	Alternative 1		Alternative 2		Alternative 2	
		Score	Weighted Score	Score	Weighted Score	Score	Weighted Score
Meet monthly demand of 5000 units	0,15	10	1,5	10	1,5	10	1,5
20% expansion space	0,15	10	1,5	10	1,5	10	1,5
Isolated areas for welding	0,15	9	1,35	8	1,2	6	0,9
Should conform to activity relationship chart	0,1	9	0,9	8	0,8	6	0,6
Optimise facility layout	0,05	6	0,3	7	0,35	8	0,4
Sufficient external interface	0,1	8	0,8	10	1	8	0,8
Logical flow of material	0,1	6	0,6	8	0,8	8	0,8
Access to roads	0,1	7	0,7	8	0,8	9	0,9
Ease of implementation or construction	0,1	9	0,9	8	0,8	7	0,7
Totals	1	74	8,55	77	8,75	72	8,1

From the evaluation above it is evident that alternative layout 2 has the highest score (77/100) and is the best one to utilise as base for the MFP. Some small changes might be made to the layout based on the evaluation and discussions above and will be discussed in the next section.

9. Final Design

Although the basic layout of alternative 2 was used for the design, many adjustments and necessary additions were made. In terms of the general layout was the finished chair storage and finished table storage combines. Similarly, the offices & reception and manual & automated welding were also combined to allow for better flow. The powder coating area was also increased in size to allow for better ventilation around the machine. The parking area was reduced in size to allow for better flow of traffic.

Some additions that were made were that more space was allocated to WIP racks and loading docks were added to the loading bays to make it easier for trucks to park and items to be loaded. Forklift parkings were also added at raw material storage and finished goods storage. The parking in the finished goods areas are to be used by both packaging and finished goods and distribution. Motor gates and security check points were also added to the internal roads before trucks and commercial vehicles enter the grounds, which are surrounded with a fence. An outdoor area next to the cafeteria was added to increase the amount of people that can take a break at once. This is especially beneficial when 20% increase in personnel is also considered.

Clear roofing panels made from polycarbonate sheets are installed on the roof in strategic places. This is done as there is little to no windows in the facility, due a focus on security and theft prevention as well as cost savings on construction and maintenance. Small orders are collected at the same warehouse as large orders, but at the overhead door closest to reception. In this way employees handling the deliveries can easily move between these two areas.

Another important note is that at the intersection next to the parking area, trucks and small order collection vehicles will have right of way and other vehicles must yield when approaching the intersection. This should however not be a major issue as very few people will enter and exit the grounds during operating hours when the trucks come.

The facility (excluding roads & undeveloped areas) has a total area of 20 444m². The building itself has an area of 13 036m² and a height 7m, resulting in a building volume of 91 252m³. The total facility surrounded by a fence has an area of 43 462m². The roads (excluding parking) takes up an additional 8638m², resulting in a total developed area of 29 082m². The MFP can be seen in figure 9.1 on the next page.

Master Facility Plan for TIPTOP TABLES

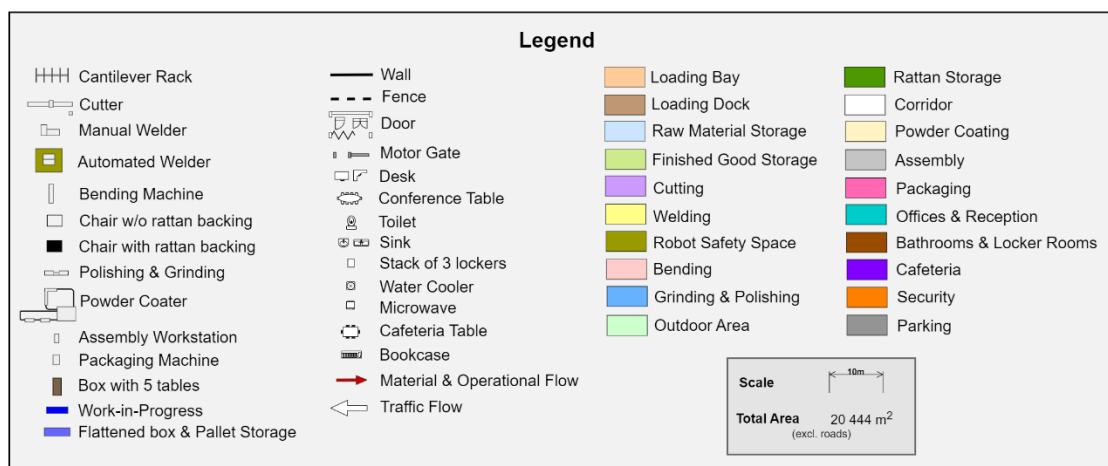
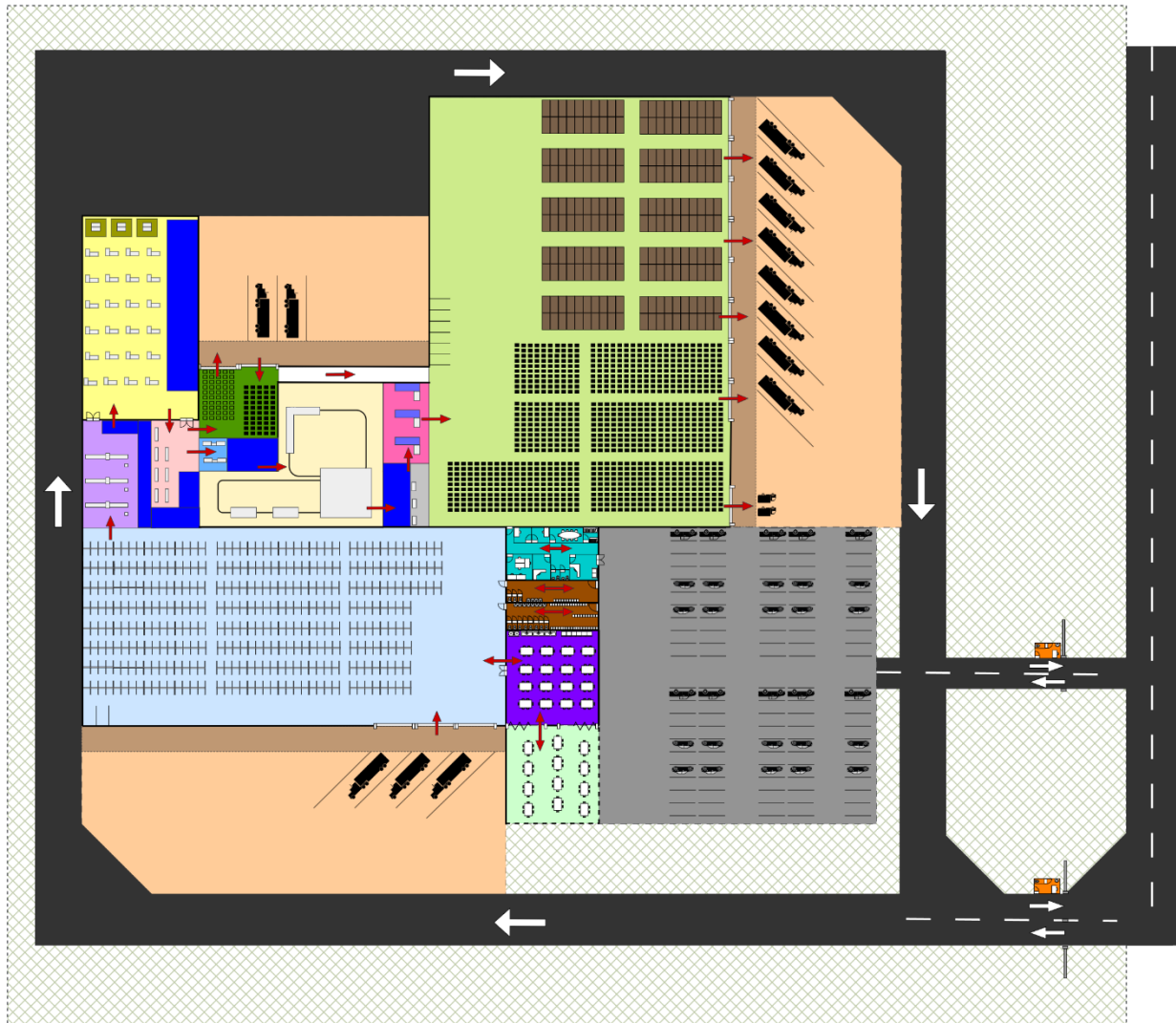


Figure 9.1: Master Facility Plan

10. Business Case

10.1. Capital Cost Estimates

The total capital cost estimates (including the equipment costs calculated above) is calculated and can be seen in table 10.2 below. A breakdown of the equipment costs can be seen in table 10.1 below. Operational and procurement costs are not taken into account as it does not form part of the facility costs.

Table 10.1: Equipment Costs

Equipment	Rate (R/equipment)	No. of equipment	Cost (R)
Cantilever Racks	15 000	88	R1 320 000
WIP Storage Racks	4 000	175	R700 000
Cutting Machine	250 000	3	R750 000
Cutoff Bins	4 000	3	R12 000
Bending Machine	60 000	7	R420 000
Manual Welder	45 000	24	R1 080 000
Automated Welder	200 000	3	R600 000
Grinder	600	2	R1 200
Polisher	350	2	R700
Powder Coater	1 200 000	1	R1 200 000
Workbenches	2 000	3	R6 000
Trolley	4 000	3	R12 000
Electric Forklifts	400 000	8	R3 200 000
Dolly with Wheels	1 500	850	R1 275 000
Packaging Machine	40 000	3	R120 000
Total			R10 696 900

Table 10.2: Capital Cost Estimates

Component	Rate (R/ m ²)	Total m ²	Cost (R)
Building	7 000	13 036	R91 252 000
Offices	10 000	112	R1 120 000
Cafeteria	10 000	210	R2 100 000
Parking & Roads	1000	10 206	R10 206 000
Restrooms	7 500	105	R787 500
Electrical	1 500	13 500	R20 250 000
Lighting	1000	13 500	R13 500 000
Ventilation	1 000	13 036	R13 036 000
Fire Safety	300	13 500	R4 050 000
Equipment			R10 696 900
Total Cost			R166 998 400

10.2. Risks and Mitigations

Table 10.3 below outlines the potential risks identified for the project along with their impacts and proposed mitigation strategies.

Table 10.3: Risks & Mitigation Strategies

Risk	Impact	Mitigation Strategy
Construction delays – Late site preparation or permitting	Project timeline slip; increased labour/equipment costs	Early engagement with local authorities for fast-track permitting; Include 10% time contingency in schedule
Cost overruns – Material price volatility (aluminium)	Budget exceedance; reduced profit margins	Lock-in key material prices via forward contracts; Maintain 10% budget contingency reserve
Equipment selection risk – Delivery or selection times longer than planned	Delayed commissioning; lost production days	Pre-qualify multiple suppliers; negotiate penalty clauses for late delivery and include delivery buffers
Labour shortages or delays in hiring skilled workers	Production can't start; lost production days and profit	Start recruitment and training early

10.3. Schedule

A high-level depiction of the project schedule can be seen below in figure 10.1.

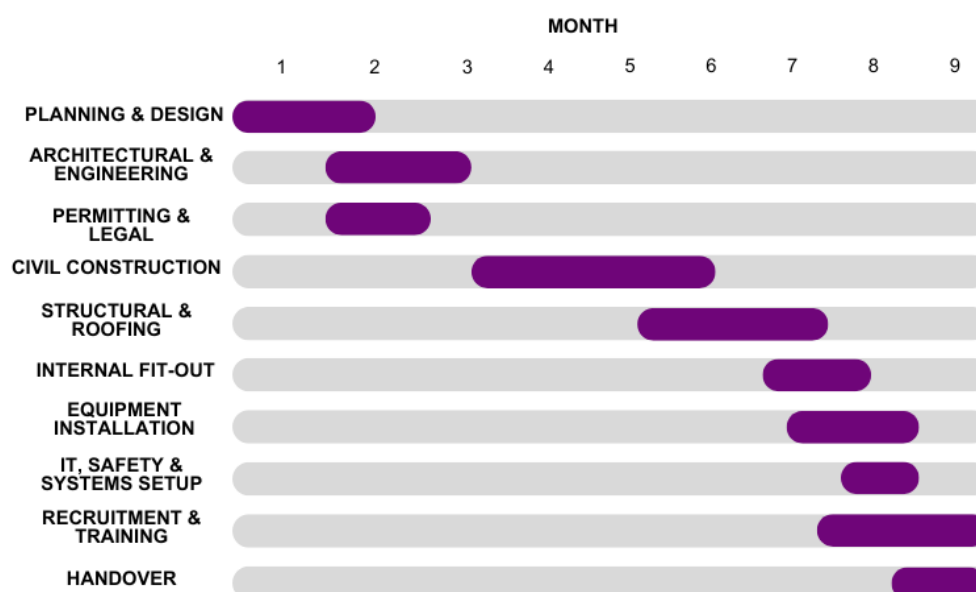


Figure 10.1: Schedule